

RegDivergence-101: An LLM Benchmark for Cross-Jurisdiction Regulatory Contradiction Detection in Life Sciences

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Pharmaceutical sponsors developing a drug for both the United States and the European Union must reconcile guidance issued independently by the FDA and the EMA. Where the two agencies require *substantively the same thing*, a sponsor can file once; where they *diverge*, a single trial design risks rejection in one region; where one agency is *silent* on a point the other regulates, the sponsor must infer obligations. Today this reconciliation is performed manually by regulatory-affairs experts. We introduce **cross-jurisdiction regulatory divergence detection**: given an FDA requirement and an EMA requirement on the same topic, classify their relationship as AGREE, DIVERGE, or SILENT. SILENT is inherently directional (SILENT_FDA vs. SILENT_EMA); we record direction per pair and report per-direction F1 alongside the collapsed label. We release **RegDivergence-101**, a 101-pair expert-grounded **pilot** evaluation benchmark (labels grounded in three peer-reviewed FDA/EMA comparison studies and primary FDA/EMA/ICH guidance text; dual-annotation inter-annotator $\kappa = 0.85$), and systematically characterise a four-method baseline hierarchy: lexical heuristic (0.511 macro-F1, 95% CI [0.411–0.605]), NLI cross-encoder (0.233), obligation-level Graph-RAG (0.663 [0.570–0.747]), and flat LLM judge / Claude Haiku (0.830 [0.747–0.908]). Three directional observations emerge at pilot scale ($n = 101$): SILENT is semantically detectable but invisible to entailment-only formulations; pair-level obligation graphs improve over lexical methods but trail flat-LLM context (CIs partially overlapping); and corpus-level graph construction is the indicated architectural target for large-scale silent-detection. RegDivergence-101 is a pilot release establishing the task formulation and baseline hierarchy; four unrepresented regulatory domains and an expansion roadmap are described in §7.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; • **Information systems** → *Document filtering*; • **Applied computing** → *Health informatics*.

Additional Key Words and Phrases: large language models, LLM benchmark, regulatory NLP, AI for compliance, FDA/EMA divergence, contradiction detection, Graph-RAG, retrieval-augmented generation, generative AI, life sciences, AGREE/DIVERGE/SILENT classification

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1

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1 Introduction

Bringing a medicine to market in both the US and EU requires satisfying two regulators that publish guidance separately and frequently *differently*. Gene-therapy sponsors face a 15-year long-term follow-up recommendation from FDA while EMA defers the duration to case-by-case risk assessment—same topic, materially different specificity. Analogous divergences span pediatric extrapolation, endpoint definitions in ulcerative colitis, biosimilar comparability, and estimand implementation. A sponsor who misses one divergence during protocol design can lose a year or more re-running a study.

Reconciling the two corpora is currently a manual, expert task. Yet the NLP building blocks for automating it all exist—document-level NLI (ContractNLI [1], DocNLI [2]), legal contradiction detection (LegalWiz [3], CLAUSE [4]), and Graph-RAG (Microsoft GraphRAG [5], GraphCompliance [6], RAGulating Compliance [7]). **No prior work combines them to detect divergence between two parallel regulatory corpora from different jurisdictions.** All existing contradiction work is *intra-document* (within one contract) or *single-jurisdiction* (contract-vs-statute).

This paper makes three contributions:

- (1) **Task formulation.** Cross-jurisdiction regulatory divergence detection as a three-way classification (AGREE / DIVERGE / SILENT) over pre-aligned FDA/EMA requirement pairs, grounded in published expert comparisons. The task is explicitly scoped to the *classification* stage given pre-aligned pairs; topic alignment is a non-trivial upstream problem addressed in §3.1 and §3.2. SILENT is treated as directional (SILENT_FDA vs. SILENT_EMA); we report per-direction F1 in Table 2.
- (2) **Pilot benchmark. RegDivergence-101**, a pilot 101-pair benchmark (59 from peer-reviewed FDA/EMA comparison studies, 42 derived from primary FDA/EMA/ICH guidance) spanning 13 topic areas, with a documented two-track pair-construction protocol, inter-annotator agreement ($\kappa = 0.85$), and bootstrap confidence intervals. Four regulatory domains are currently unrepresented; §7 describes the expansion roadmap.
- (3) **Baseline hierarchy with three directional observations.** Four methods systematically compared at pilot scale ($n = 101$), yielding directional empirical observations on the task’s structure and the path to a corpus-level solution. Observations are reported with bootstrap 95% CIs; confirmatory significance testing awaits a larger benchmark.

2 Related Work

Document-level NLI. ContractNLI [1] introduced document-level entailment/contradiction/not-mentioned classification with evidence extraction, and showed that contradiction detection lags entailment, especially under “negation by exception.” DocNLI [2] extended NLI to full documents.

Legal contradiction detection (2025–2026). LegalWiz [3] uses a multi-agent generation framework for legal contradiction detection but is confined to single-document regimes. Better Call CLAUSE [4] benchmarks 7,500+ perturbed contracts across ten anomaly categories and finds that LLMs miss subtle errors and struggle to *justify* them legally. Contradiction Detection in RAG Systems [8] uses LLMs as context validators. Two recent RAG-faithfulness studies bear directly on our LLM-judge and graph-RAG results: Chen et al. [9] probe whether a RAG system can recognise when retrieved context is incorrect, and Qian et al. [10] show that the surface relevance of retrieved evidence does not guarantee it *warrants* a claim—the same failure mode documented in our Observation 2, where modal-register similarity (RECOMMENDED vs. MANDATORY) misleads the pair-level graph.

Graph-RAG and regulatory compliance. Microsoft GraphRAG [5] builds entity graphs with community summaries for local and global queries. GraphCompliance [6] aligns *policy graphs* (regulatory requirements) with *context graphs* (organisational facts) for GDPR; it explicitly lists “handling conflicting regulatory interpretations” as open. RAGulating Compliance [7] builds an ontology-free regulatory knowledge graph for traceable QA.

Cross-jurisdiction regulatory benchmarks. The closest concurrent work is **Sino-US-DrugQA** [11] (11,871 MCQA items, US–CN), which shares our concept-alignment thesis but cannot express the SILENT class. Comparative-regulatory-science surveys document FDA/EMA divergence across dosing [12], oncology labelling [13], and multi-country SmPCs [14], but as human-authored comparisons, not ML benchmarks. RegDivergence-101 is the first FDA/EMA benchmark with an explicit SILENT class.

Foundation models and AI governance beyond text. The momentum of foundation models now spans multimodal scientific modelling [15], vector-graphics generation [16] and animation [17], and vision-language navigation [18], while parameter-efficient domain adaptation and ensembling of small language models [19] lower deployment cost. In high-stakes sectors such as healthcare, deep-learning deployment is increasingly governed by policy-aligned evaluation frameworks that weigh safety and regulatory constraints over raw accuracy [20]. RegDivergence-101 extends this momentum to cross-jurisdiction regulatory reasoning, a high-stakes text domain that remains underrepresented in current foundation-model evaluation.

Gap. Contradiction detection is intra-document or single-jurisdiction; Graph-RAG targets single-corpus QA or single-jurisdiction compliance; FDA/EMA divergence is documented manually but never automated. Our task sits precisely in this gap.

3 Task and Dataset

3.1 Task Definition

For a topic t , let f_t be an FDA requirement statement and e_t the corresponding EMA statement. The label y_t is:

- **AGREE** — both jurisdictions require substantively the same thing (same modal strength, same threshold, same scope of applicability). Paraphrase equivalence and synonymous modals (*shall* / *must*) are treated as AGREE; minor drafting differences that do not alter sponsor obligations are disregarded.

- **DIVERGE** — the requirements directly conflict: they impose different thresholds, different modal strengths (*shall* vs. *should*), different timelines, or mutually incompatible design specifications.
- **SILENT** — one jurisdiction does not address what the other regulates. SILENT is *directional* by construction:
 - **SILENT_FDA**: FDA is silent on this topic; an EMA requirement exists but no corresponding FDA requirement was found in the full published FDA corpus (guidance documents, Q&As, MaPPs).
 - **SILENT_EMA**: EMA is silent on this topic; an FDA requirement exists but no corresponding EMA requirement was found in the full published EMA corpus (guidelines, Q&As, reflection papers).

RegDivergence-101 records direction in the `silent_direction` field and reports per-direction F1 in Table 2. A unified **SILENT** label ($= \text{SILENT_FDA} \cup \text{SILENT_EMA}$) is retained in Table 1 for backward compatibility.

Epistemic status of SILENT labels. A SILENT label is an absence claim: it asserts that no counterpart requirement was found after an exhaustive corpus search. This carries irreducible epistemic uncertainty. We bound this uncertainty through (a) a documented corpus-search protocol (§3.3) and (b) a per-pair `doc_confidence` flag (HIGH / MODERATE). Downstream work should treat MODERATE-confidence SILENT pairs as soft labels.

Task scope and assumptions. This formulation assumes requirement pairs are *pre-aligned*—both the FDA and EMA sides address the same regulatory topic. Topic alignment is a non-trivial upstream problem: FDA and EMA guidance frequently address overlapping domains at inconsistent levels of specificity, making it unclear whether two requirements are “on the same topic” at all. RegDivergence-101 is constructed from expert-panel comparisons that *already provide aligned pairs* and from a two-track alignment procedure described in §3.2; we treat topic alignment as a separate, upstream task and refer readers to §7 for the corpus-level architecture that would encompass it.

3.2 Pair Construction Protocol

RegDivergence-101 is constructed via two tracks that differ in how topical correspondence between FDA and EMA requirements is established. Per-pair source-track attribution (**track**: A or B) is released with the dataset.

Track A — expert-comparison-derived pairs (59 pairs, 58.4%). All 59 pairs were extracted from three open-access peer-reviewed studies whose explicit purpose was comparative mapping of FDA and EMA requirements in a defined therapeutic domain:

- (1) Vieujean et al. [21] — ulcerative colitis endpoints and trial design (40 pairs).
- (2) Boesen et al. [22] — psychiatric drug-trial guidelines (10 pairs).
- (3) Schwarz and Decristoforo [23] — radiopharmaceutical nonclinical requirements (9 pairs).

In each source paper, domain experts had already established co-topicality through structured literature mapping or side-by-side dossier comparison. Our annotation task for Track A was to classify the *relationship* (AGREE / DIVERGE / SILENT) between requirements that the source authors had already deemed co-topical. We inherit, document, and cite their alignment judgements; per-pair source-study attribution is released with the dataset.

Track B — primary guidance pairs (42 pairs, 41.6%). For pairs derived from ICH E9(R1), E11A [24], and FDA/EMA biosimilar and gene-therapy guidance, topical alignment was established in two steps:

- (1) *Anchor identification.* Requirements were associated with ICH Common Technical Document (CTD) module and sub-topic designations (e.g., Module 5.3.5.1 for comparative bioavailability study reports). Requirements sharing the same CTD anchor were candidate pairs.
- (2) *Expert confirmation.* The primary annotator independently reviewed each candidate pair and confirmed or rejected co-topicality before the pair entered the dataset.

Specificity exclusion criterion. A candidate Track B pair was excluded if the FDA and EMA requirements operated at different levels of the regulatory hierarchy (e.g., one requirement addressing trial design at the *statistical analysis plan* level while the other addressed the same concept at the level of *general statistical principles*). Such pairs would conflate regulatory level-of-detail differences with substantive DIVERGE labels. A small number of candidate pairs were excluded on this basis (the count and representative cases are documented in the released dataset’s `excluded_pairs.jsonl` file). The complete exclusion-criteria checklist is released with the dataset so third parties can audit or extend our alignment judgements.

Each pair records `{id, topic, track, source, fda_text, ema_text, label, silent_direction, doc_confidence, disputed, annotation_notes}`. The combined release contains **101 pairs**: 42 AGREE, 38 DIVERGE, 21 SILENT, across 13 topic areas. We report macro-F1 to weight all classes equally.

3.3 Annotation Protocol

Annotators. Annotator A (C. Wu) is a solution architect for a regulatory-document AI platform; Annotator B (Z. Zhou) is a product manager for an AI-powered clinical-trial management platform. Neither holds formal RA credentialing; Track A expert-panel papers serve as the external human-expert anchor.

Stages. Stage 1: Wu assigned labels using a definitions-based framing (Track A labels inferred from source authors’ stated assessments; Track B labels assigned against full guidance text). Stage 2: Zhou independently re-labelled all 101 pairs under a sponsor-obligation framing to probe framing-independent stability. Both annotators are co-authors; the κ is a cross-framing consistency measure, not identical-instruction IAA. An independent third-annotator pass is planned for RegDivergence-500.

IAA. Human cross-framing $\kappa = 0.85$ (3-class, 101 pairs; observed agreement 0.90)—*substantial* per Landis and Koch [25], on par with MedNLI (0.65) and ContractNLI (≈ 0.72). SILENT is near-perfect ($\kappa = 0.98$); AGREE/DIVERGE more variable (≈ 0.79 – 0.81). A secondary LLM prompt-robustness check on a 36-pair sample yielded $\kappa = 0.542$ (LLM test-retest, not human IAA).

Disagreements (n = 10). Six resolved by source-paper language; four genuinely borderline pairs retain the Stage 1 label and are flagged `disputed = True` in the dataset with written justifications.

SILENT confidence. All 21 SILENT pairs carry `doc_confidence = MODERATE`: exhaustive corpus search was performed (full FDA and EMA guidance, Q&As, reflection papers) but absence claims remain epistemically soft. The HIGH/MODERATE audit is deferred to RegDivergence-500.

4 Baselines

4.1 Evaluation Context

RegDivergence-101 is an **evaluation benchmark**, not a training dataset. All four baselines are zero-shot or few-shot. Adjacent methods are separated by $\Delta = 0.15\text{--}0.60$ macro-F1 (min: Graph-RAG vs. LLM, $\Delta = 0.167$; CIs partially overlapping); bootstrap CI half-widths $\approx \pm 0.08\text{--}0.10$ at $n = 101$. Results are reported as a ranking trend, not pairwise significance.

4.2 Shared Evaluation Protocol

We evaluate four methods under a shared reporting framework: **stratified 5-fold splits** averaged over 5 random seeds, with **bootstrap 95% CI** (1,000 resamples). The protocol differs by method class. The lexical heuristic and NLI cross-encoder have tunable thresholds and use **nested** CV (inner folds tune thresholds, outer folds report). The Graph-RAG classifier and LLM judge have no tunable hyperparameters: the Graph-RAG classifier is reported over the same outer folds, and the LLM judge is scored directly on all 101 pairs. CIs for both LLM methods are bootstrap-only.

4.3 Lexical Heuristic

No neural model, no API—fully offline and deterministic. Combines TF-IDF cosine similarity with five conflict signals: negation asymmetry, divergence cue pairs (e.g., *single-arm* vs. *randomised*), modal-strength asymmetry (*must/shall* vs. *may/encouraged*), numeric-threshold mismatch, and one-sided specificity (concrete number vs. case-by-case deferral). A shared ICH citation is a strong AGREE signal.

4.4 NLI Cross-Encoder

`typeform/distilbert-base-uncased-mnli`, run in both directions; labels derived from contradiction/neutral/entailment probabilities with CV-tuned thresholds. We use DistilBERT-MNLI as a representative off-the-shelf NLI system. The structural SILENT gap observed in §6 (Observation 1) is model-agnostic: any NLI model lacking an explicit SILENT output class will map absence-of-regulation to *neutral*, collapsing SILENT recall. Substituting DeBERTa-v3-large or a modern cross-encoder would improve AGREE/DIVERGE F1 but would not resolve the SILENT structural gap without task-specific supervision.

4.5 Graph-RAG Pair-Level Classifier

For each text, Claude Haiku extracts a policy triplet $\langle \text{subject}, \text{obligation_level}, \text{requirement}, \text{conditions} \rangle$ where $\text{obligation_level} \in \{\text{MANDATORY}, \text{PROHIBITED}, \text{RECOMMENDED}, \text{PERMITTED}, \text{SILENT}\}$. Only $\text{MANDATORY} \leftrightarrow \text{PROHIBITED}$ is a hard-coded DIVERGE rule; all other pairs are escalated to a second LLM call that reasons over the *obligation structure* of the aligned node pair, with explicit calibration that modal-register differences (RECOMMENDED vs. MANDATORY) do not automatically imply DIVERGE.

Within-model scope. Both the Graph-RAG classifier and the flat LLM judge use `claude-haiku-4-5-20251001`. The comparison between them is therefore a *within-model ablation*—with vs. without explicit obligation-graph structure—not an architectural comparison across model families. Conclusions about the relative value of graph structure should be confirmed with a second model family before generalisation.

Table 1. Macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for the four baselines ($n = 101$ pairs).

Method	Macro-F1 [95% CI]	Acc	AGREE F1 [CI]	DIVERGE F1 [CI]	SILENT F1 [CI]
Lexical heuristic ¹	0.511 [0.411–0.605]	0.541	0.718 [0.605–0.809]	0.466 [0.320–0.593]	0.334 [0.133–0.537]
NLI cross-encoder ²	0.233 [0.152–0.303]	0.238	0.279 [0.103–0.449]	0.383 [0.247–0.506]	0.025 [0.000–0.086]
Graph-RAG pair-level ³	0.663 [0.570–0.747]	0.673	0.754 [0.646–0.851]	0.586 [0.433–0.722]	0.640 [0.478–0.771]
LLM judge (Haiku)⁴	0.830 [0.747–0.908]	0.842	0.905 [0.831–0.965]	0.805 [0.696–0.892]	0.778 [0.609–0.914]

¹Nested 5-fold CV, 5 seeds; thresholds tuned on inner folds only. ²Same CV protocol, thresholds tuned on inner folds. ³No tunable hyperparameters; bootstrap CI only. ⁴claude-haiku-4-5-20251001, default temperature; bootstrap CI only. Estimated cost: ~\$0.10 for all 101 pairs.

4.6 LLM Judge (Flat Pairwise)

Claude Haiku with explicit AGREE / DIVERGE / SILENT definitions, prompted once per pair with both texts. No graph, no retrieval. Full prompt and code at Appendix A.

5 Results

Table 1 reports macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for all four methods.

Table 2. Per-direction SILENT F1 for all four methods (bootstrap 95% CI, $n = 101$ pairs). SILENT.FDA ($n = 13$): FDA is silent, EMA regulates. SILENT.EMA ($n = 8$): EMA is silent, FDA regulates.

Method	SILENT.FDA F1 [95% CI]	SILENT.EMA F1 [95% CI]
Lexical heuristic	0.277 [0.000–0.545]	0.393 [0.000–0.706]
NLI cross-encoder	0.049 [0.000–0.167]	0.000 [0.000–0.000]
Graph-RAG pair-level	0.716 [0.516–0.882]	0.535 [0.267–0.762]
LLM judge (Haiku)	0.795 [0.588–0.952]	0.738 [0.429–0.952]

SILENT-FDA: 13 pairs (FDA silent, EMA active). SILENT-EMA: 8 pairs (EMA silent, FDA active). Wide CIs on SILENT-EMA reflect the small stratum ($n = 8$). Bootstrap 1,000 resamples; stratified by 3-class label.

Point estimates increase monotonically from lexical to Graph-RAG to the LLM judge. At $n = 101$ the marginal bootstrap CIs partially overlap, so we report these as a consistent ranking trend rather than a claim of statistical significance; a paired significance test is left to future work. **Lexical ceiling.** Characteristic errors share the signature of *high lexical overlap, opposed regulatory stance*: FDA “modified Mayo score of 5–9” vs. EMA “full Mayo score of 9–12” (same vocabulary, different instrument); FDA 15-year LTF vs. EMA case-by-case (same topic, different specificity). Resolving these requires representing what each requirement *asserts*, not surface similarity.

6 Analysis

6.1 What the Baseline Hierarchy Reveals

Observation 1 — SILENT is semantically detectable but invisible to NLI framing. SILENT F1 = 0.025 under NLI (no native SILENT class) vs. 0.778 under the LLM judge (explicit definition). This contrast is large and consistent across bootstrap resamples; it reflects a structural constraint—the

SILENT class requires a formulation that can express absence, which NLI entailment cannot—rather than a model-capacity effect. We note this observation is based on a distilbert-class model; a higher-capacity NLI encoder would improve AGREE/DIVERGE F1 but would not resolve the absence-detection gap without task-specific supervision (§6).

Observation 2 — For Claude Haiku on this dataset, pair-level obligation graphs improve over lexical (+0.15 F1) but trail flat prompting (−0.17 F1); CIs partially overlap at pilot scale. The graph makes the obligation structure explicit (MANDATORY / RECOMMENDED / etc.), which helps distinguish clear stance differences from similar-sounding text. But it discards full-text context that flat prompting uses to separate *same-requirement* / *different-modal-register* (AGREE) from *same-topic* / *different-threshold* (DIVERGE). Pairs like gene-01 (FDA 15 years vs. EMA case-by-case) both extract as RECOMMENDED and the graph calls them AGREE; the flat LLM correctly identifies the threshold conflict from the text [10]. *Scope note.* Both the graph classifier and the flat LLM judge use Claude Haiku; this comparison is a within-model ablation (with vs. without explicit graph structure) rather than a general claim about graph methods vs. LLM methods. Replication with a second model family is left to future work.

Observation 3 — Two distinct problem scopes suggest two distinct architectural targets. The *pair-level scope* — given pre-aligned sentences, label AGREE/DIVERGE/SILENT — is largely addressed by the flat LLM judge (0.830), approaching a task ceiling. The *corpus-level scope* — “is EMA silent on this topic *anywhere* across its guideline corpus?” — requires finding aligned pairs, not just classifying them; a pairwise LLM cannot address this, but a corpus-level policy graph can. These are different tasks. This paper establishes baselines for the pair-level scope; §7 motivates the corpus-level architecture as the next contribution. This observation is architectural and directional; it is not a validated finding.

6.2 Source-Stratified Performance Analysis

To assess whether performance reflects source-study-specific patterns rather than genuine task competence, we analyse performance stratified by track and by source concentration.

Track-level. On AGREE/DIVERGE classes, the LLM judge scores *lower* on Track A (0.769/0.745) than Track B (0.966/0.923): exposure to source studies does not inflate performance. The macro-F1 gap (Track B 0.630 vs. Track A 0.765) is explained by SILENT support ($n = 0$ vs. $n = 21$), not memorisation.

Source-concentration. 40/101 pairs come from one source (Vieujean et al.). LLM judge on the JCC stratum ($n = 40$): 0.804; on all other pairs ($n = 61$): 0.844. Performance is marginally *higher* outside JCC, ruling out domain-concentration inflation.

7 Future Work

Corpus-level policy graph. The correct long-term target is corpus-level silent detection and divergence discovery: (1) build per-jurisdiction policy graphs indexing all FDA/EMA guidance at corpus scale; (2) align requirement nodes via entity normalisation and embedding similarity; (3) classify aligned pairs as AGREE/DIVERGE/SILENT by obligation-structure comparison; (4) generate natural-language explanations. This architecture subsumes the pair-alignment assumption: alignment becomes a graph-construction step, not a pre-processing input.

RegDivergence-500. Four domains are unrepresented in the pilot: CMC, adaptive trial design/estimates, statistical analysis plan requirements, and labelling/SmPC. RegDivergence-500 (≈ 500 pairs) will cover all four via a community-annotation pipeline seeded from a ~ 45 -source FDA/EMA comparison inventory. Annotation guidelines are released with this paper. A third jurisdiction (PMDA) via FDA/EMA/PMDA concordance studies [12, 13] will add an ≈ 40 -pair FDA $\times$ PMDA subset. Multi-model replication and an independent third-annotator validation pass are planned. A cross-pair generalisation probe against Sino-US-DrugQA [11] will be the first test of whether conclusions transfer beyond the FDA/EMA pair.

8 Limitations

Topic alignment. Track B co-topicality judgements are single-annotator; Track A inherits source-study alignment methodology. Reproducibility depends on the released exclusion-criteria checklist ([data/alignment_protocol.md](#)).

Scale and coverage. 101 pairs from three source studies limits statistical power (directional observations, not confirmed findings). Oncology, rare diseases, medical devices, and vaccines are absent. RegDivergence-500 (§7) is required before domain-general conclusions.

SILENT labels. All 21 SILENT pairs carry `doc_confidence = MODERATE`; treat them as soft labels. The pair-level SILENT task is a proxy for the correct corpus-level formulation (§7).

Single-model evaluation. All LLM-dependent components (triplet extraction, graph classification, flat judge, annotation check) use Claude Haiku. The Graph-RAG vs. LLM comparison is a within-model ablation, not a cross-model comparison. The $\kappa = 0.542$ LLM consistency check reflects the same model’s label tendencies. Multi-model replication is deferred to RegDivergence-500.

Contamination. Haiku may have seen FDA/EMA guidance during pretraining. The source-stratified analysis (§6.2) provides evidence against inflation; future work should evaluate on post-cutoff guidance.

Use norms. RegDivergence-101 is a pilot benchmark: not sufficient for deployment decisions, not domain-generalizable without held-out evaluation, and not a sole criterion for AI procurement in regulatory-affairs workflows. Conclusions are specific to the FDA/EMA pair until cross-pair generalisation probes are run.

9 Conclusion

We introduced cross-jurisdiction regulatory divergence detection, released **RegDivergence-101**, a 101-pair expert-grounded **pilot** evaluation benchmark ($\kappa = 0.85$, documented two-track pair-construction protocol, per-direction SILENT annotation), and characterised a four-method baseline hierarchy with bootstrap CIs. Three directional observations at pilot scale ($n = 101$, CIs partially overlapping for adjacent methods): SILENT requires explicit absence-aware formulation (NLI F1 0.03 $\rightarrow$ LLM F1 0.78); for Claude Haiku on this dataset, pair-level graph structure improves over lexical heuristics (+0.15 F1) but trails flat prompting (−0.17 F1); and the two-scope distinction (pair-level vs. corpus-level) motivates the corpus-level graph architecture as the next contribution. The flat LLM judge (0.830) establishes a pair-level performance reference at pilot scale; the corpus-level architecture proposed in §7 is the correct target for large-scale silence discovery. The benchmark is a pilot: four regulatory domains (CMC, clinical trial design, statistical analysis, labelling) remain unrepresented, and the planned RegDivergence-500 extension with expanded dual-annotation will establish the task at production scale.

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This work was conducted independently. Benchmark pairs were derived from publicly available FDA/EMA guidance documents and open-access peer-reviewed comparison studies. LLM API calls used the Anthropic API (Claude Haiku, `claude-haiku-4-5-20251001`).

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A Prompts and Code

Full prompts (LLM judge system/user, Graph-RAG Stage 1 triplet-extraction, Stage 2 graph-classification), evaluation scripts, and dataset schema are available at the project repository: <https://github.com/chuchugo/cross-jurisdiction-regulatory-divergence>

LLM judge (`llm_judge.py`): system prompt defines AGREE/DIVERGE/SILENT with explicit absence framing; user prompt pairs `{fda_text}` and `{ema_text}`. Model: `claude-haiku-4-5-20251001`, temperature 1.0, max tokens 100.

Graph-RAG (`graph_rag.py`): Stage 1 extracts a policy node `{subject, obligation_level, requirement, conditions}`; Stage 2 classifies the aligned node pair. Hard rules: MANDATORY $\leftrightarrow$ PROHIBITED $\Rightarrow$ DIVERGE (no LLM call); both SILENT $\Rightarrow$ SILENT. Max tokens: 300.